CHAPTER 11

WORLD CLIMATE AND CLIMATE CHANGE

CHAPTER ESSENCE AT A GLANCE:
Master Vladimir Koppen's empirical climatic classification scheme, temperature and precipitation thresholds across groups A, B, C, D, E, and H, greenhouse radiative forcing, and natural versus anthropogenic drivers of global climate change.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Koppen Empirical Classification System:** Vladimir Koppen developed an empirical scheme (1918) relating monthly temperature and precipitation data to the geographic boundaries of major world vegetation belts.

• **Major Climatic Groups:** Five thermal groups: A (Tropical humid, coldest month above 18 degrees Celsius), C (Warm temperate), D (Cold snow forest), E (Polar ice and tundra), and H (Highland); and one moisture-deficient group B (Dry arid and semi-arid climates).

• **Greenhouse Effect and Anthropogenic Warming:** Gases including carbon dioxide, methane, nitrous oxide, and chlorofluorocarbons absorb outgoing longwave infrared radiation, trapping heat and raising global surface mean temperatures.

• **Drivers of Climate Variability:** Natural drivers include Milankovitch orbital cycles (eccentricity, obliquity, precession), solar sunspot variations, and volcanic aerosols; magnified exponentially by fossil fuel combustion and industrial emissions.

ESSENTIAL GEOGRAPHICAL MODELS & LANDFORMS

$$R_{\text{threshold}} = 20T + \{0 \text{ (winter rain)}, \quad 140 \text{ (uniform)}, \quad 280 \text{ (summer rain)}\}$$

Koppen Aridity and Dry Boundary Mathematical Threshold

$$\Delta F = 5.35 \times \ln\left(\frac{C}{C_0}\right) \rightarrow \text{Radiative Greenhouse Forcing (W m}^{-2}\text{)}$$

Logarithmic Radiative Climate Forcing Equation

$$\text{Milankovitch Astronomical Cycles} \rightarrow \text{Eccentricity (100 kyr)} + \text{Obliquity (41 kyr)} + \text{Precession (23 kyr)}$$

Orbital Pacemakers Driving Glacial and Interglacial Cycles

$$\text{Koppen Humid Tropical Taxa} \rightarrow \{Af \text{ (Rainforest)}, \quad Am \text{ (Monsoon)}, \quad Aw \text{ (Savanna)}\}$$

Group A Seasonal Precipitation Subdivisions

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In Koppen's scheme, capital letters designate major temperature groups (A, C, D, E) or dryness (B), whereas lowercase letters indicate precipitation seasonality (f = no dry season, m = monsoon, w = dry winter, s = dry summer).

• **Alert:** Group B (Dry climates) is unique because it is classified on the basis of moisture deficiency and high evapotranspiration rather than thermal temperature boundaries.

• **Alert:** Never confuse weather (instantaneous day-to-day atmospheric conditions) with climate (aggregate average weather conditions over a standard minimum period of 30 years).

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. What were the primary criteria used by Vladimir Koppen in his empirical classification of world climates? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Criteria:** Mean monthly and annual values of temperature and precipitation. **[1 Mark]**

• **Vegetation Relationship:** Selected specific threshold values based on the geographic distribution of distinct natural vegetation formations. **[1 Mark]**

2. What do the Koppen climatic symbols 'Af', 'Am', and 'Aw' represent? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Af:** Tropical Wet Climate (Equatorial rainforest) with no dry season throughout the year. **[0.75 Mark]**

• **Am:** Tropical Monsoon Climate with a short dry winter season compensated by heavy monsoon rains. **[0.75 Mark]**

• **Aw:** Tropical Wet and Dry Climate (Savanna) with distinct dry winter season. **[0.5 Mark]**

3. Distinguish between 'BWh' (Hot Desert) and 'BWk' (Cold Desert) climates with global examples. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **BWh (Hot Desert Climate):** Characterized by extremely high summer temperatures exceeding 40 degrees Celsius and mean annual temperature above 18 degrees Celsius (e.g. Sahara, Thar, and Arabian Deserts). **[1.5 Marks]**

• **BWk (Cold Desert Climate):** Located in continental interiors at mid-latitudes with cold freezing winters and mean annual temperature below 18 degrees Celsius (e.g. Gobi Desert, Patagonia, and Ladakh). **[1.5 Marks]**

4. Explain the major greenhouse gases responsible for global warming and state their primary anthropogenic sources. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Carbon Dioxide (CO2):** The primary driver, emitted through fossil fuel combustion (coal, oil, gas) in power plants and vehicles, and deforestation. **[1 Mark]**

• **Methane (CH4):** Emitted from paddy fields, livestock enteric fermentation, and anaerobic decomposition in landfills. **[1 Mark]**

• **Chlorofluorocarbons and Nitrous Oxide:** Emitted from industrial refrigeration, aerosol propellants, and chemical nitrogen fertilizers. **[1 Mark]**

5. Discuss the major causes of global climate change, distinguishing between natural astronomical causes and anthropogenic activities. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Astronomical Causes (Milankovitch Cycles):** Variations in the shape of Earth's orbit (eccentricity), tilt of Earth's axis (obliquity between 22.1 and 24.5 degrees), and wobble of Earth's axis (precession) alter seasonal insolation. **[1.5 Marks]**

• **Other Natural Causes:** Sunspot cycles causing solar output oscillations, and massive volcanic eruptions injecting sulfur dioxide aerosols that block sunlight and cause global cooling. **[1 Mark]**

• **Anthropogenic Causes:** Rapid industrialization, rampant burning of fossil fuels, mass destruction of tropical rainforests, and expansion of emission-intensive agriculture amplifying the greenhouse effect. **[1.5 Marks]**

6. Describe the unique characteristics and global distribution of the Mediterranean Climate (Cs). [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Unique Precipitation Regime:** Characterized by warm, dry, rainless summers and mild, wet, rainy winters due to seasonal shifting of wind belts. **[1.5 Marks]**

• **Shifting Wind Mechanism:** In summer, subtropical high-pressure dry trade winds dominate; in winter, the belt shifts equatorward bringing rain-bearing moist Westerlies. **[1.5 Marks]**

• **Distribution:** Found along western margins of continents between 30 and 45 degrees latitude: Mediterranean coastal basin, central California, central Chile, Cape Province of South Africa, and south-western Australia. **[1 Mark]**